Appendix I

Mark–Houwink Parameters for Homopolymers

Polymer	Solvent	Temp (°C)	$K \times 10^2$ (ml/g)	a	MW range ($M \times 10^{-3}$)	Re-mark	Ref
Polystyrene	THF	20	1.55	0.692			1
		23	0.682	0.766	50–1000	A	2
		25	1.11	0.725		B	3
			1.60	0.706	> 3		4
			1.17	0.725		B	5
			1.17	0.717		B	6
			1.25	0.707			7
			1.68	0.69			8
			1.41	0.70			9
			1.25	0.717			10
			1.42	0.70			11
			1.16	0.724			12
			1.28	0.70			13
			0.609	0.768			14
		30	1.76	0.679			15
			0.86	0.74		B	16
			1.05	0.731	$36–10^4$		17
			1.57	0.698			18
			1.05	0.731		B	19
			1.62	0.701		B	20
			1.112	0.723			21
			1.28	0.712	$> 10^4$	C	22
			17.1	0.428	$< 10^4$		
		31.15	1.25	0.717	4–1800		23
		35	1.258	0.715			24
			1.580	0.704			25
			1.60	0.703			26
		38	1.247	0.720			27
		40	1.90	0.68			28
			1.31	0.719	19.1–1110	B	29
			1.51	0.706	5.11–1030	A	
	benzene	25	1.13	0.73			30
		30	1.623	0.695	5–400		31
	toluene	25	1.05	0.73			32
			0.662	0.76			33
			11.4	0.470	< 9000		34
			1.06	0.728	> 9000		

Polymer	Solvent	Temp (°C)	$K \times 10^2$ (ml/g)	a	MW range ($M \times 10^{-3}$)	Re-mark	Ref
		30	1.786	0.680	5–2000		31
			1.20	0.714	17–10^4		17
			9.86	0.499	0.56–17		
			1.57	0.695		B	35
		60	0.797	0.75			36
		68	1.528	0.691			15
	chloroform	25	0.716	0.76	100–2800		37
		30	1.36	0.708	24–10^4		17
			11.1	0.502	0.16–24		
			0.49	0.796			38
	p-dioxane	30	2.241	0.655	5–100		31
	MC	25	1.533	0.695			39
		30	1.157	0.718			40
	MC/DCA	30	1.4	0.70		D	41
	cyclohexane	34.5	8.3	0.50		B	35
		44.5	2.30	0.608		B	
	cyclohexanone	40	3.02	0.643		E	42
	MEK	30	1.79	0.642		B	35
	DMF	20	2.40	0.63			43
		30	2.606	0.612	5–400		31
	DMF + 0.01 M LiBr	60	1.30	0.662			44
	DMF	135	2.80	0.606			45
	OCP	25	14.3	0.488			46
		90	1.2	0.70			28
	TCB	127	1.21	0.71			47
		130	0.895	0.727		A	2
		135	1.21	0.707			48
			1.72	0.67			49
		140	1.90	0.655			50
			1.34	0.70			51
		145	4.14	0.610			52
			3.53	0.617			
			1.75	0.67			53
	ODCB	87	1.0	0.73		B	54
		135	1.156	0.715			22
			1.38	0.70	2–900		55
			4.87	0.606			56
			1.51	0.693			57
		138	1.38	0.7		F	58
	m-cresol	130	0.846	0.715		B	59
		135	2.02	0.65	4–200		28
	cis-decalin	25	4.0	0.574	20–24000	B	60
	HFIP/ toluene (2:8)	50	1.795	0.657			61
	Phenol/TCB (1:1)	135	1.303	0.69			62
	CNP	208	1.86	0.657			63
PMS	THF	25	4.2	0.608			11
	TCB	135	1.61	0.672			48

Polymer	Solvent	Temp (°C)	$K \times 10^2$ (ml/g)	a	MW range ($M \times 10^{-3}$)	Re-mark	Ref
PVC	THF	25	1.63	0.766	20–170		4
			1.50	0.77	20–100	B	6
			7.2	0.61	27–100		8
			4.48	0.70			14
			1.6	0.77		G	64
			4.76	0.673			65
		30	1.60	0.77	100–1000	B	5
		35	1.556	0.690			24
			1.8	0.76		B	66
	TCB	110	1.61	0.762			67
PVAc	THF	25	3.50	0.63	10–1000	B	5
			0.942	0.737			12
			0.51	0.791			68
			1.6	0.70			69
		30	1.49	0.698			70
		35	1.56	0.708			71
	TCB	135	1.37	0.687			70
PMMA	THF	23	0.93	0.69			8
		25	1.28	0.69	150–1200		9
			0.958	0.695			13
			0.787	0.75			65
			1.48	0.677			72
			1.99	0.660			73
		35	1.22	0.69			26
	chloroform	25	1.47	0.714			73
		30	0.43	0.80			38
	acetone	25	3.66	0.578			73
	MEK	25	0.71	0.72			38
	benzene	25	0.55	0.76			30
			4.38	0.597			73
		30	0.52	0.76			74
	toluene	25	1.91	0.68			33
			2.08	0.64	> 7800		34
	cyclohexanone	40	3.66	0.603		E	42
	DMF	25	0.404	0.787		A	75
			2.5	0.625		B	
	TFE	30	0.281	0.88			76
(100% isotactic)	THF	25	1.665	0.660			77
(92% iso)			1.690	0.659			
(83% iso)			1.719	0.658			
(35% iso)			0.984	0.692			
(23% iso)			0.9577	0.695			
(15% iso)			0.8892	0.694			
PEMA	THF	25	1.549	0.679			14
			2.25	0.66			78
PBMA	THF	25	0.503	0.758			14
	MEK	25	0.613	0.7258	10^4–2×10^4		79
			0.970	0.68	10^5–7×10^5		80

Polymer	Solvent	Temp (°C)	$K \times 10^2$ (ml/g)	a	MW range ($M \times 10^{-3}$)	Re-mark	Ref
POMA	THF	25	5.556	0.560			14
PLMA	THF	25	0.73	0.69			81
PSMA	THF	30	0.90	0.67		B	82
PMA	THF	25	0.388	0.82			83
	toluene	35	2.1	0.60			84
PtBA	THF	25	0.33	0.80		H	85
Polyethylene	ODCB	135	4.77	0.70	6–700		56
			4.90	0.74			86
		138	5.06	0.7		F	58
	TCB	135	5.50	0.704			87
			4.34	0.724			48
			9.54	0.64			49
			5.3	0.7		F	88
HDPE	ODCB	135	5.05	0.693	10–1000		57
	TCB	127	3.58	0.74			47
		130	3.92	0.725			89
		135	9.54	0.64			90
			5.26	0.70			88
			5.23	0.70			91
			7.1	0.67			92
			5.63	0.70			93
		140	3.95	0.726			50
			3.23	0.735			94
		145	1.73	0.784			52
	decalin	135	4.17	0.71			95
			3.9	0.74			96
			4.6	0.73			91
		138	6.2	0.7		F	97
	decane	120	6.98	0.65			95
LDPE	ODCB	135	5.06	0.70	0.2–200		99
LLDPE	TCB	145	5.91	0.69			53
Hydrogenated PB	TCB	135	2.7	0.746	2.4–40		58
PP	TCB	135	1.37	0.75			48
			1.76	0.73			91
		145	1.56	0.76			53
	ODCB	135	1.0	0.78			86
			2.42	0.707		B	98
			1.10	0.80			99
	decalin	135	1.10	0.80			86, 91
Polyisobutylene	THF	25	5	0.60		H	100
		40	6.47	0.593	0.73–67.6	A	29
			5.79	0.593	0.82–80.4	B	
Polycarbonate	THF	25	3.99	0.77		I	28
			4.9	0.67	7–270		101
			4.12	0.69	10–300	J	102
		35	3.18	0.680		K	25
PB	toluene	30	5.19	0.679	$0.5–10^4$		17
			2.888	0.734			21
		68	2.103	0.753			15

Polymer	Solvent	Temp (°C)	$K \times 10^2$ (ml/g)	a	MW range ($M \times 10^{-3}$)	Re-mark	Ref
	THF	25	1.012	0.621			27
			18.4	0.58			65
			1.6	0.776			67
		30	4.57	0.693	$0.5-10^4$		17
			2.56	0.74		B	82
		40	3.96	0.697		L	103
	chloroform	30	4.51	0.704	$0.19-10^4$		17
		40	5.78	0.67	10–100		28
(*cis*/*trans* = 0.8)		30	76.0	0.44			
(20% *cis*-20% vinyl)			25	2.36	0.75	3–6	
(8% vinyl)			4.57	0.693	80–1100		10
(28% vinyl)			4.51	0.693	20–300		
(52% vinyl)			4.28	0.693	20–200		
(73% vinyl)			4.03	0.693	20–200		
SBR(25% styrene)	THF	25	4.10	0.693	2.4–40		
		40	3.18	0.70	70–1000		28
Natural rubber (poly-*cis*-isoprene)	THF	25	1.09	0.79	10–1000	G	
Polyisoprene	THF	25	1.77	0.753	40–500		10
Butyl rubber	THF	25	0.85	0.75	4–4000		28
PDSX	toluene	25	0.828	0.72			38
		60	0.977	0.725			36
	chloroform	30	0.54	0.77			38
	cyclohexane	35	0.63	0.77			104
	ODCB	87	8.19	0.50	20–800		54
		138	3.83	0.57	25–300		55
	DMAC	25	1.3	0.69		M	105
Nylon 6	*m*-cresol	25	32	0.62	0.5–5		28
	OCP	90	6.2	0.64			
	HFB	25	4.2	0.76			106
	TFE	25	4.58	0.742	20–100		107
			5.36	0.75	13–100		108
Nylon 66	*m*-cresol	25	3.53	0.79	0.15–50		28
		130	4.01	1.00	8–24	B	59
	HFIP		19.8	0.63			109
Nylon 610	*m*-cresol	25	1.35	0.96	8–24		28
Nylon 11	*m*-cresol	30	9.1	0.69			110
Nylon 12	*m*-cresol	25	4.6	0.75	10–125		28
	HFIP/ toluene (2:8)	50	6.163	0.6585			61
Trogamid T	DMF	22	2.737	0.706			111
TFA-nylon 6	MC	25	3.355	0.64		N	112
	THF		1.66	0.70		O	113
TFA-nylon 12	MC	30	0.738	0.81		P	114
TFA-Trogamid	THF	25	0.87	0.73		O	113
PET	*m*-cresol	25	0.077	0.95			28
		135	1.75	0.81	27–32		
			2.0	0.90	0.45–0.8		
	OCP	23	6.31	0.658			115
	PTCE	23	7.44	0.648			

Polymer	Solvent	Temp (°C)	$K \times 10^2$ (ml/g)	a	MW range ($M \times 10^{-3}$)	Re-mark	Ref
	HFIP	23	5.20	0.695			
	PFP	23	3.85	0.723			
	HFIP/PFP (1:1)	23	4.50	0.705			
		25	6.31	0.658			46
	MC/DCAA	30	4.5	0.68		Q	41
	MC/HFIP (7:3)	25	4.034	0.691		R	116
LCA polyester	PFP	50	0.9205	1.05			117
PTHF	cyclohexanone	40	4.16	0.676		E	42
Polyoxymethylene	DMF	135	0.932	0.79			45
PVDBC	THF	25	9.1	0.827			118
PVB	THF	25	1.4	0.80		H	119
POA	THF	25	0.842	0.202	> 12.5	G	120
			0.961	0.674	< 12.5	G	
PDEPP	THF	25	2.5	0.65		S	121
PDPPP, PAPP	THF	30	1.19	0.649		B, T	20
PVK	THF	25	0.127	0.80		U	122
PPS	CNP	210	0.891	0.747			63
PEEK	sulfuric acid	25	0.3849	0.94		B	62
	phenol/TCB (1:1)	115	7.588	0.67		B	62

Abbreviations. THF – tetrahydrofuran; MC – methylene chloride; MEK – methyl ethyl ketone; DMF – *N,N*-dimethylformamide; OCP – *o*-chlorophenol; TCB – 1, 2,4-trichlorobenzene; ODCB – *o*-dichlrobenzene; HFIP – hexafluoroisopropanol; CNP – 1-chloronaphthalene; TFE – trifluoroethanol; DMAC – *N,N*-dimethylacetamide; HFB – hexafluorobutanol; PTCE – phenol/tetrachloroethane (6:4); PFP – pentafluorophenol; PMS – poly(methyl styrene); PVC – poly(vinyl chloride); PVAc – poly(vinyl acetate); PMMA – poly(methyl methacrylate); PEMA – poly(ethyl methacrylate); PBMA – poly(butyl methacrylate); POMA – poly(octyl methacrylate); PLMA – poly-(*n*-lauryl methacrylate); PSMA poly(stearyl methacrylate); PMA – poly(methyl acrylate); PtBA – poly-(*tert*-butyl acrylate); HDPE – high density polyethylene (PE); LDPE – low density PE; LLDPE – linear LDPE; PP – polypropylene; PB – polybutadiene; SBR – styrene-butadiene rubber; PDSX – polydimethylsiloxane; Trogamid T – poly(trimethylhexamethylene terephthalamide); PET – poly(ethylene terephthalate); LCA polyester-liquid crystalline aromatic polyester; PTHF – poly(tetrahydrofuran); PVDBC – poly(*N*-vinyl-3,6-dibromocarbazole); PVB – poly(vinyl butyral); POA – poly[*N*-(*n*-octadecyl) maleimide]; PDEPP – poly(diethoxyphospazene); PDPPP – poly(diphenyl phosphazene); PAPP – poly-(aryloxy phosphazen); PVK – poly(*N*-vinyl-3,6-dibromocarbazole); PPS – poly(phenylene sulfide); PEEK – poly(arylether ether ketone).

Remarks. A – $[\eta] = KM_n^a$; B – $[\eta] = KM_w^a$; C – narrow MW distribution; D – MC/dichloroacetic acid (8:2) + 0.01 M tetrabutylammonium acetate; E – 0.01 M tetrabutylammonium nitrate included; F – $[\eta] = KM_v^a$; G – use K and a from [4] for PS; H – use K and a from [6] for PS; I – 4,4′-dihydroxy-2,2-propane carbonate; J – use K and a from [9] for PS; K – bisphenol A/diethylene glycol (50:50); L – 35% cis, 55% trans, 10% 1,2; M – poly(methyl) (pyridine-3-yl) siloxane; N – use K and a from [37] for PS; O – use $K = 1.31 \times 10^{-2}$ and $a = 0.714$ for PS; P – use K and a from [38] for PS; Q – MC/dichloroacetic acid + 0.01 M tetrabutylammonium acetate; R – use $K = 7.998 \times 10^{-2}$ and $a = 0.54$ for PS; S – 0.1% tetra(*n*-butylammonium) bromide included; T – 0.1 M LiBr included; U – use K and a from [3] for PS.

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Appendix II

Mark–Houwink Parameters for Polar Polymers

Polymer	Solvent	Temp (°C)	$K \times 10^2$ (ml/g)	a	MW range ($M \times 10^{-3}$)	Remark	Ref
PAN	DMF	20	3.25	0.725			1
		25	2.43	0.75			2
		60	0.146	0.897		A	3
			2.12	0.75		A	4
PVP	DMF	60	1.85	0.646		A	3
	$MeOH/H_2O$ (1:1)	25	0.851	0.72		B	5
P2VP	THF	25	2.23	0.66			6
		35	3.02	0.61			7
	TFA + $NaNO_3$	35	0.575	0.922		C	
			1.62	0.745		D	
			2.57	0.651		E	
	pyridine	25	0.445	0.78		F	8
	NMP	80	0.88	0.73		G	
	DMF	50	1.23	0.69		H	
	sodium formate	25	0.25	0.93		I	9
			0.50	0.83		J	
			0.17	0.95		K	
	see Remark	25	2.5	0.68		L	10
Polyacrylamide	water	25	6.80	0.66			11
			1.0	0.755	40–9000	M	12
		30	0.631	0.80			13
	0.1 M Na_2SO_4	25	1.94	0.70			14
	0.5 M NaCl	25	1.21	0.746		pH 9, M	15
	1 M NaCl		1.84	0.72		pH 9, M	
NaPA	0.3 N NaCl	25	1.69	0.75			16
Phenol (RN)	acetone	25	4.7	1.90	1–8, M_n		17
		30	63.1	0.28	0.37–28, M_n		18
	THF	30	73	0.28	0.54–4.7, M_n		
	DMF	25	18.0	0.51	1–8, M_n		17
	1N NaOH		16.6	0.48			
(HO)	acetone	30	8.13	0.5	0.69–2.6, M_n		18
PVPDMAEMA	$LiNO_3$	25	1.42	0.67		N	19
PTAC 50	1 M NaCl	25	5.64	0.90		O	20
PTAC 100	1 M NaCl	25	1.55	0.97		O	20
PTMAC 50	1 M NaCl	25	185	0.66		O	20
PTMAC 100	1 M NaCl	25	59.8	0.72		O	20
PMVEMA	tris-buffer	35	3.17	0.70		P	21
PDMDAAC	0.1 M NaCl	30	3.5	0.62		O	22

Polymer	Solvent	Temp (°C)	$K \times 10^2$ (ml/g)	a	MW range ($M \times 10^{-3}$)	Re-mark	Ref
Poly(N-vinylacetamide)							
	H_2O	30	16	0.52		Q	22
Polyamic acid	DMAc	30	2.01	0.79		R	23
PEG, PEO	H_2O	25	1.25	0.78			24
		30	1.25	0.78			13
	$LiNO_3$	25	2.8	0.72		N	19
		35	3.48	0.70		P	21
PEG	0.1 N $NaNO_3$	35	10.1	0.58			25
PEO			3.47	0.70			
	0.42 N NaOH	25	2.5	0.66			26
	p-dioxane	25	3.50	0.71	0.06–19		27
	DMF	25	2.40	0.73	1–30		
		60	5.0	0.653		A	3
			5.50	0.643		A	4
	$MeOH/H_2O$ (1:1)	25	2.21	0.75		B	5
PEG	$MeCN/H_2O$ (2:8)	35	87.3	0.60			25
PEO			2.45	0.73			
	$MeOH/H_2O$ (2:8)	25	3.4	0.75			28
PPG	benzene	30	3.241	0.686	1–4		27
	toluene	30	0.447	0.920	0.8–4		
	p-dioxane	30	2.247	0.715	1–4		
	DMF	30	5.284	0.598	0.8–4		
PVA	H_2O	25	14.0	0.60	10–70		29
	0.1N $NaNO_3$	35	13.3	0.56	13–250		30
Cellulose nitrate	THF	25	25.0	1.00	95–2300		31
			82	1.0	DP < 1	S	32, 33
			446	0.76	DP > 1	S	
			6.44	0.73		T	34
Nitrocellulose	THF	25	3.21	0.83			35
Methyl cellulose	H_2O	25	10.7	0.61		O	36
Dextran	H_2O	25	9.78	0.5			24
	0.1 M NaCl	23	15.7	0.451		U	37
		25	13.6	0.47			38
	0.1 M $NaNO_3$	25	37	0.40		V	39
	0.2 M $NaNO_3$		48.5	0.39			40
	0.2 M $NaNO_3$ + 0.025 M KH_2PO_4	30	47.6	0.34			41
	0.42 N NaOH	25	12.0	0.48			26
	NaOH (0.005 N)	25	18.0	0.438			42
	(0.0185 N)		18.5	0.431			
	(0.097 N)		15.8	0.451			
	(0.287 N)		12.4	0.474			
	(0.501 N)		13.2	0.478			
	(0.671 N)		14.7	0.465			
	(0.877 N)		15.1	0.464			
Branched dextran	$MeOH/H_2O$		103	0.25			43
Pullulan	0.1 M NaCl	25	1.79	0.67			38
	0.2 M $NaNO_3$		4.22	0.64			40
	0.1 M $NaNO_3$ + 0.025 M KH_2PO_4	30	1.70	0.72			41

Polymer	Solvent	Temp (°C)	$K \times 10^2$ (ml/g)	a	MW range ($M \times 10^{-3}$)	Re-mark	Ref
	NaAc buffer	25	1.79	0.67			44
	H_2O	25	2.23	0.67		Q	45
			2.36	0.66			46
Polysaccharide	0.1 N $NaNO_3$	35	2.14	0.66			25
	$MeCN/H_2O$ (2:8)	35	3.72	0.59			
	$MeOH/H_2O$ (2:8)	25	4.4	0.6			28
	H_2O	25	6.9	0.6			
Xanthan	0.1 M $NaNO_3$	25	0.024	1.10		V	39
Xylan	0.5 M NaOH	25	2.67	0.73			47
NaPSS	phosphate	25	0.60	0.74		Q, W	45
		25	0.50	0.80		Q, X	45
	NaCl (0.005 N)	25	0.057	1.061			42
	(0.05 N)		0.106	0.935			
	(0.20 N)		0.209	0.845			
	(0.50 N)		0.285	0.794			
	(1.0 N)		0.416	0.741			
	0.1 M NaCl	60	0.24	0.80			48
	NaOH (0.005 N)	25	0.038	1.079			42
	(0.0185 N)		0.146	0.935			
	(0.097 N)		0.458	0.805			
	(0.287 N)		0.281	0.815			
	(0.501 N)		0.198	0.828			
	(0.671 N)		0.174	0.830			
	(0.877 N)		0.110	0.851			
	0.42 N NaOH	25	0.2	0.82			26
	see Remark		1.09	0.74		Y	10
Chitosan	see Remark	25	19.9	0.59		O, Z	49
HA	0.15 M NaCl	25	6.54	1.16			50
	see Remark		3.6	0.78		Z2	51
Ch4-S	0.2 M NaCl	25	17	1.01			50
	0.02 M NaCl		3.1	0.74			52
P(DL)LA	THF	31.15	5.49	0.639			53
Proteins	guanidine		3.16	0.66		Z3	54
	see Remark	25	0.0174	1.09		Z4	55

Abbreviations. DMF – *N,N*-dimethylformamide; THF – tetrahydrofuran; TFA – trifluoroacetic acid; NaAc – sodium acetate; phosphate – phosphate buffer; NMP – *N*-methylpyrrolidone; PAN – poly(acrylonitrile); PVP – poly(vinyl pyrrolidone); P2VP – poly(2-vinyl pyridine); NaPA – sodium polyacrylate; phenol (RN) – phenol-formaldehyde resin (random novolak resin); Phenol(HO) – phenol-formaldehyde resin (high-ortho resin); PEG – poly(ethylene gylcol); PEO – poly(ethylene oxide); PPG – poly(propylene glycol); PVA – poly(vinyl alcohol); NaPSS – sodium poly(styrene sulfonate); PVPDMAEMA – poly(vinylpyrrolidone-co-dimethylaminoethyl methacrylate); PTAC50 – poly(acrylamide-co-*N,N,N*-trimethylammonium ethyl acrylate chloride); PTAC100 – poly(*N,N,N*-trimethylammonium ethyl acrylate chloride); PTMAC50 – poly(acrylamide-co-*N,N,N*-trimethylammonium ethyl methacrylate chloride); PTMAC100 – poly(*N,N,N*-trimethylammonium ethyl methacrylate chloride); PMVEMA – poly(methyl vinyl ether-co-maleic anhydride); PDMDAAC – poly(dimethyldiallylammonium chloride); HA – hyaluronic acid; Ch4-S – chondroitin 4-sulfate; P(DL)LA – poly(DL-lactic acid).

Remarks. A – 0.01 M LiBr included; B – 0.01 M $LiNO_3$ included; C – 0.1 %TFA + 0.01 N $NaNO_3$; D – 0.1 %TFA + 0.20 N $NaNO_3$; E – 9.1 %TFA + 0.50 N $NaNO_3$; F – use $K = 0.379 \times 10^{-2}$ and $a = 0.81$ for PS; G – use $K = 1.8 \times 10^{-2}$ and $a = 0.68$ for PS; H – use $K = 2.88 \times 10^{-2}$ and $a = 0.60$ for PS; I – 0.1 M sodium formate at pH 3.50; J – 0.5 M sodium formate at pH 3.50; K – 0.1 M sodium formate + 20 % methanol at pH 3.55; L – 80 %(0.5 M Na_2SO_4 + 0.5 M MeCOOH) + 20 % MeCN; M – $[\eta] = KM_v^a$; N – 0.5 M $LiNO_3$ (pH 7.0); O – $[\eta] = KM_w^a$; P – 0.1 M tris (hydroxymethyl amino methane) with 0.2 M $LiNO_3$ adjusted to pH 9.0 with HNO_3; Q – $[\eta] = KM_n^a$; R – 0.03 M LiBr + 0.03 M H_3PO_4 + 1 vol %THF in dimethyl acetamide, use $K = 1.21 \times 10^{-2}$ and $a = 0.69$ for PS; S – $[\eta] = K(\mathrm{DP})^a$, DP – degree of polymerization; T – 0.01 M acetic acid included; U – 0.02 M disodium hydrogen phosphate at pH 7 included; V – 0.2 g/l NaN_3 + 0.1 %ethylene glycol included; W – 0.51 M phosphate buffer (pH 8.0); X – 0.05 M phosphate buffer (pH 8.0); Y – 80 % of 0.03M Na_2SO_4 + 20 % MeCN; Z – 0.5 M acetic acid + 0.5 M sodium acetate; Z2 – 0.1 M phosphate + 0.1 M NaCl; Z3 – 6 M guanidinium hydrochloride; Z4 – 0.1 M sodium phosphate + 2 % sodium dodecyl sulfate + 0.02 M dithiothreitol (pH 7.0), proteins (bovine serum albumin, chymotrypsinogen, lysozyme chloride, and cytochrome C).

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Appendix III

Mark–Houwink Parameters for Copolymers

Polymer	Solvent	Temp (°C)	$K \times 10^2$ (ml/g)	a	Remark	Ref
PVC-PVAc	THF	25	6.72	0.611	10–13% VAc	1
EPM	toluene	30	3.051	0.76	50% PP, A	2
EPDM	THF	35	27.4	0.54		3
EPR	ODCB	135	4.43	0.725		4
			2.42	0.707		5
	toluene	30	3.1	0.72	62% PE, B	6
E-NORB	xylene	90	10.0	0.535	50%	7
	DEB	120	4.93	0.589		
EVA	THF	20	9.7	0.62	27–29% EVA, C	8
P (S-MMA)	THF	ns	0.775	0.76	alternating	9
	MEK		1.15	0.69		
	toluene		1.09	0.73		
	n-chlorobutane		1.08	0.70		
	THF		0.641	0.76	1:1 block	
	MEK		1.466	0.67		
	toluene		1.318	0.69		
	n-chlorobutane		4.446	0.55		
	toluene	ns	1.14	0.70	statistical, 30% S	10
	n-chlorobutane		2.65	0.60		
	toluene		1.32	0.71	57% S	
	n-chlorobutane		2.49	0.63		
	toluene		0.832	0.75	71% S	
	n-chlorobutane		1.76	0.67		
P (S-MAH)	THF	25	3.98	0.595	MAH 5- 50mol%, B	11
P (S-AN)	DMF	20	1.45	0.71	12% S, D	12
			1.80	0.71	25% S	
			2.65	0.72	52% S	
		25	1.86	0.690	15–82% AN, E	13
	THF		1.98	0.681	0–35% AN, E	
	MEK		3.30	0.619	10–48% AN, E	
P (AN-EVE)	DMF	55	0.142	0.9636	18% EVE, F	14
P (SA-AA)	0.5M NaCl	25	0.81	0.789	10% SA, G, pH 9	15
	1M NaCl		0.307	0.860		
	0.5M NaCl		1.09	0.775	20% SA	
	1M NaCl		1.307	0.759		
	0.5M NaCl		5.78	0.655	50% SA	
	1M NaCl		2.98	0.695		
	0.5M NaCl		0.638	0.844	70% SA	
	1M NaCl		0.026	1.072		

Abbreviations. THF – tetrahydrofuran; ODCB – *o*-dichlorobenzene; DEB – diethylbenzene; MEK – methyl ethyl ketone; ns – not specified; PVC – polyvinyl chloride; PVAc – polyvinyl acetate; EPM – ethylene–propylene copolymer; EPDM – terpolymer of ethylene, propylene, diene monomers; EPR – ethylene-propylene rubber; E-NORB – ethylene-norboarnene (50%); EVA – ethylene–vinyl acetate copolymer; S – polystyrene (PS); MMA – polymethyl methacrylate (PMMA); MAH – maleic anhydride; AN – polyacrylonitrile (PAN); EVE – ethyl vinyl ether; SA – sodium acylate; AA – acrylamide.

Remarks. A – use $K = 1.112 \times 10^{-2}$ and $a = 0.723$ for PS; B – $[\eta] = KM_w^a$; C – use $K = 1.55 \times 10^{-2}$ and $a = 0.692$ for PS; D – use $K = 2.40 \times 10^{-2}$ and $a = 0.63$ for PS and $K = 3.25 \times 10^{-2}$ and $a = 0.725$ for PAN; E – $[\eta] = K(W_{AN} + 1.041)^a M_w^a$, W_{AN}: weight fraction of AN unit; F – 0.05M LiBr included, use $K = 2.2 \times 10^{-2}$ and $a = 0.615$ for PS; G – $[\eta] = KM_v^a$; H – use $K = 1.17 \times 10^{-2}$ and $a = 0.717$ for PS.

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Appendix IV

Specific Refractive Index Increments

Sample	Solvent	Wave-length (nm)	dn/dc (ml/g)	Temp (°C)	A_2 ($\times 10^3$)	Ref, Remark
Polystyrene	toluene	632.8	0.110	23		1
		632.8	0.108	25		2, A
		632.8	0.108			3
		632.8	0.107			4
		632.8	0.109	22	0.5–0.3	5
		632.8	0.111	20		6
		670	0.108			7
		589.3	0.107			8, C
		546.0	0.104			9
	benzene	546.0	0.1064			9
		546.0	0.109	25		10
		436	0.112	25		10
	CCl_4	546.0	0.146			9
	tetrahydrofuran (THF)	488	0.199			10
		632.8	0.190			3
		632.8	0.193	20		6
		632.8	0.1845	25		11
		632.8	0.184	30		12
		632.8	0.186			4
		632.8	0.184			13
		632.8	0.188	22	0.47–0.3	5
		632.8	0.192	25		14
		632.8	0.184	40		15
		670	0.180	25		11
		670	0.1805	30		16, D
	chloroform	632.8	0.155			3
		632.8	0.1608			17
		546.0	0.169	25		10
		436	0.161	25		10
	acetone	670	0.224			7
	methyl ethyl ketone (MEK)	632.8	0.212	22	0.125–0.06	5
		632.8	0.214			7
		670	0.213			7
		589.3	0.212			8, C
		546.0	0.220			9
	tetrahydronaphthalene	670	0.0615			7

Sample	Solvent	Wave-length (nm)	dn/dc (ml/g)	Temp (°C)	A_2 ($\times 10^3$)	Ref, Remark
Polystyrene	MC/DCAA (8:2)	632.8	0.156			18, E
	ethyl benzoate	632.8	0.103	15		19
	dimethyl-	632.8	0.165	20		7
	formamide (DMF)	632.8	0.159			20, F
	tetrachloroethylene (TCE)	632.8	0.0934	25		21
	p-dioxane	546.0	0.152	35		22
(star)		546.0	0.168	35		22
	cyclohexane	546.0	0.170	35		22
(star)		546.0	0.168	35		22
Poly(dimethylsiloxane)	TCE	632.8	−0.0932	25		21
	toluene	632.8	−0.0913	26		20
		632.8	−0.0648	26		20
Poly(styrene-divinylbenzene)	DMF/LiBr	546	0.170	30		23
	THF	546.1	0.1963	30		24
Poly(styrene-methyl methacrylate)	ethyl benzoate	632.8	0.0398	15		19, G
Poly(methyl methacrylate)	THF	632.8	0.083	25	0.2	25
		632.8	0.084			26, H
		632.8	0.084			13
		632.8	0.084			14
		632.8	0.084			27
		632.8	0.09			28
		589	0.0865	23		29
		546	0.0871	25		10
		436	0.0887	25		10
	HFIP	632.8	0.190	25		30, J
		632.8	0.190			27
	toluene	546	0.0157	25		10
		488	0.010	18		19
		439	0.0094	25		10
	DMF	632.8	0.053	25		31
	MEK	546	0.1140			9
		546	0.1173	25		10
		436	0.100	25		10
	benzene	546.0	−0.010			9
		546.0	−0.011	25		10
		436	−0.0039	25		10
	CCl_4	546.0	0.023			9
	acetone	546.0	0.1340			9
Poly(ethyl methacrylate)	MEK	546.0	0.104	23		10
	ethyl acetate	436	0.109	25		10
Poly(*n*-butyl methacrylate)	MEK	546.0	0.102	25		10
		436	0.103	25		10
	benzene	546.0	−0.014	25		10
		436	−0.023	25		10

Sample	Solvent	Wave-length (nm)	dn/dc (ml/g)	Temp (°C)	A_2 (× 10^3)	Ref, Remark
Poly(methacrylic acid)	ethanol	436	0.154	25		10
	methanol	546	0.134	25		10
Poly(methyl acrylate)	benzene	436	−0.019	30		10
	acetonitrile	546.0	0.118	25		10
		436	0.120	25		10
Poly(ethyl acrylate)	acetone	546.0	0.107	23		10
		436	0.109	23		10
	chloroform	546.0	0.0363	25		10
	MEK	546.0	0.0856	20		10
		436	0.0870	20		10
Poly(*n*-butyl acrylate)	acetone	436	0.0870	20		10
	benzene	546	−0.0292	30		10
	THF	546	0.0651	30		10
		632.8	0.067			6
	toluene	546	−0.0239	30		10
		632.8	−0.024	20		6
	MEK	632.8	0.097	20		6
	DMF	632.8	0.016	20		6
Poly(*tert*-butyl acrylate)	MEK	546	0.0818	25		10
	propyl alcohol	546	0.074			32
Poly(vinyl chloride)	THF	632.8	0.101			20
		589.0	0.106	23		29
		589.0	0.115			29
		589.0	0.115	25		10
		578	0.1010	16.5		10
		546.0	0.1017	16.5		10
		546.0	0.116	25		10
		435.8	0.115	20		33
		435.8	0.1124	20		34
		435.8	0.1074	25		35
		435.8	0.104	16.5		10
		435.8	0.119	25		10
	cyclohexanone	632.8	0.0650			20
		589.0	0.076	25		10
		546.0	0.077	25		10
		435.8	0.079	25		10
Polyisoprene	toluene	632.8	0.031			3
	THF	632.8	0.127			3
		589.0	0.156			29
	chloroform	632.8	0.093			3
(*cis*-1,4)	benzene	546.0	0.0194	20		10
		436	0.0143	20		10
(*trans*-1,4)	cyclohexane	436	0.117			10
Polybutadiene	toluene	632.8	0.032			3
	THF	632.8	0.132			3
		632.8	0.1295	26		20, K
		589.0	0.132	40		29
	chloroform	632.8	0.094			3
		546.0	0.083			10

Sample	Solvent	Wave-length (nm)	dn/dc (ml/g)	Temp (°C)	A_2 (× 10^3)	Ref, Remark
Polybutadiene						
(*cis*-1,4)	cyclohexane	632.8	0.1151	30		36
(*trans*-1,4)		632.8	0.1084	30		36
(1,2-)		632.8	0.0873	30		36
		632.8	0.111	30		36, L
		546	0.118			10
		546	0.1174	25		9
		436	0.126			10
		436	0.1345	25		9
	benzene	546	0.0086	20		10
		546	0.0099	25		9
		436	0.0117	20		10
		436	0.0160	25		9
	hexane	546	0.1523	25		9
		436	0.2181	25		9
Polychloroprene	THF	632.8	0.1274	25		11
	MEK	546	0.156	25		10
		436	0.162	25		10
Polyisobutylene	1-chloronaphthalene	632.8	−0.122	135		37
		578	−0.128	135		37
		546	−0.131	135		37
		436	−0.146	135		37
	trichlorobenzene	632.8	−0.051	135		37
	(TCB)	578	−0.053	135		37
		546	−0.054	135		37
		436	−0.066	135		37
	THF	632.8	0.1026	23		20, M
		632.8	0.125	25	0.35	38
		546.1	0.114	25		39
	toluene	632.8	0.106	25		2, B
Poly(1-butene)	1-chloronaphthalene	632.8	−0.177	135		37
		578	−0.183	135		37
		546	−0.187	135		37
		436	−0.206	135		37
	TCB	632.8	−0.103	135		37
		578	−0.105	135		37
		546	−0.108	135		37
		436	−0.120	135		37
	n-nonane	436	−0.108	35		10
Polyethylene	TCB	632.8	−0.097	135		20
		632.8	−0.104	135		12
		632.8	−0.098	145		12
		632.8	−0.098	145		40
		632.8	−0.091	145		41
		546	−0.106	135		10
(MW 11000)		589.3	−0.1063	135		8, C
(MW 18400)		589.3	−0.1085			8, C
(MW 11000)	1-chloronaphthalene	632.8	−0.177	145		20

Sample	Solvent	Wave-length (nm)	d*n*/d*c* (ml/g)	Temp (°C)	A_2 ($\times 10^3$)	Ref, Remark
Polyethylene		589.3	−0.1883			8
		546.0	−0.191	135		42
	decane	577	0.1200	125		43
		546	0.1213	125		43
		436	0.1263	125		43
		404	0.1233	125		43
	o-dichlorobenzene	632.8	−0.056	135		20
	(ODCB)	589	−0.090	135		29
		546	−0.078	120		10
	1-cyclonaphthalene	546	−0.214	125		10
		436	−0.245	125		10
HDPE	TCB	632.8	−0.112	145	2.00	44
		632.8	−0.107	135		37
		578	−0.108	135		37
		546	−0.110	135		37
		436	−0.125	135		37
(SRM 1475)		632.8	−0.104	135	1.90	45
	ODCB	632.8	−0.056	145	1.41	44
		589	−0.069	135		29
(SRM 1475)		632.8	−0.056	135		45
	1-chloronaphthalene	632.8	−0.189	145	0.68	44
		632.8	−0.183	135		37
		578	−0.188	135		37
		546	−0.192	135		37
		436	−0.215	135		37
(SRM 1475)		632.8	−0.177	145		45
(SRM 1475)		589.3	−0.1932	135		8
	diphenylmethane	546	−0.125	142		46
LDPE	TCB	632.8	−0.108	145	3.05	44
		632.8	−0.102	135		37
		578	−0.103	135		37
		546	−0.105	135		37
		436	−0.117	135		37
(SRM 1476)		632.8	−0.091	135	4.3	45
		632.8	−0.091	140	0.42	47
	ODCB	632.8	−0.051	145	1.71	44
		589.0	−0.104	138		29
	1-chloronaphthalene	632.8	−0.185	145	0.87	44
		632.8	−0.178	135		37
		578	−0.184	135		37
		546	−0.189	135		37
		546	−0.195	125		48
		436	−0.212	135		37
(SRM 1476)		589.3	−0.1916	135		8
Linear-LDPE	TCB	632.8	−0.106	145	1.50	44
		632.8	−0.106	145		49, N
		632.8	−0.103	145		49, O
	ODCB	632.8	−0.048	145	1.05	44
	1-chloronaphthalene	632.8	−0.180	145	0.79	44

Sample	Solvent	Wave-length (nm)	dn/dc (ml/g)	Temp (°C)	A_2 ($\times 10^3$)	Ref, Remark
Polypropylene	1-chloronaphthalene	546	−0.227	125		10
		546	−0.191	125		27
		436	−0.228	125		10
(isotactic)		632.8	−0.173	135		37
(isotactic)		578	−0.180	135		37
(isotactic)		546	−0.184	135		37
(isotactic)		436	−0.205	135		37
(atactic)		632.8	−0.177	135		37
(atactic)		578	−0.183	135		37
(atactic)		546	−0.188	135		37
(atactic)		436	−0.211	135		37
	TCB	632.8	−0.095	135		50
		632.8	−0.093	145		49
		632.8	−0.091	145		20
(MW 235800)		632.8	−0.092	145		51
(MW 175100)		632.8	−0.094	145		51
(isotactic)		632.8	−0.102	135		37
(isotactic)		578	−0.105	135		37
(isotactic)		546	−0.107	135		37
(isotactic)		436	−0.121	135		37
(atactic)		632.8	−0.105	135		37
(atactic)		578	−0.109	135		37
(atactic)		546	−0.111	135		37
(atactic)		436	−0.123	135		37
Polyoxymethylene	DMF	632.8	−0.0505	135		52
Polyethylene terephthalate	HFIP	632.8	0.257		2.9	53
	HFIP/PFP (5:5)	632.8	0.242			53, P
	MC/DCAA (8:2)	632.8	0.148	RT		18, E
	DCAA	589	0.106			54, Q
	m-cresol	589.0	−0.073	25		29
	o-chlorophenol	589.0	−0.070	25		29
Polyamide 6	HFIP	632.8	0.2375	25		55
	MC/DCAA (8:2)	632.8	0.180	RT		56
	o-chlorophenol	589.0	−0.012	90		29
Polyamide 11	MC/DCAA (8:2)	632.8	0.133	RT		56
	HFIP	546	0.25	30	7.8−5.6	57
Polyamide 12	MC/DCAA (8:2)	632.8	0.133	RT		56
	HFIP	632.8	0.2212			58
		546	0.25	30		59
Polyamide 6, 6	MC/DCAA (8:2)	632.8	0.182	RT		56
(46K)			0.188	RT		56
(32K)			0.194	RT		56
	HFIP	632.8	0.241	25	4.8	60
	formic acid	632.8	0.136	25		60
	m-cresol	589.0	−0.016	130		29
Polyamide 6, 9	MC/DCAA (8:2)	632.8	0.163	RT		56

Sample	Solvent	Wave-length (nm)	dn/dc (ml/g)	Temp (°C)	A_2 ($\times 10^3$)	Ref, Remark
Polyamide 6, 10	MC/DCAA (8:2)	632.8	0.163	RT		56
Polyamide 6, 12	MC/DCAA (8:2)	632.8	0.149	RT		56
Polyamide 6T	MC/DCAA (8:2)	632.8	0.201	RT		56
Silk (Silkworm)	HFIP	632.8	0.236			61
(Spider)	HFIP	632.8	0.235			61
Polyacrylonitrile	DMF	632.8	0.092	25		31
		546	0.082	20		10
		436	0.087	20		10
	dimethylsulfoxide	546	0.031	25		10
		436	0.029	25		10
	dimethylacetamide	546	0.0769	25		10
		436	0.0767	25		10
Poly(acrylonitrile-styrene) (28.5/71.5)	THF	632.8	0.1546	23		20
Polyurethane	THF	632.8	0.0863	25		62
		632.8	0.1469	23		20, R
Poly(methyl vinyl ether)	cyclohexane	632.8	0.050	50		63
PMVEMA	0.1 M tris	632.8	0.227			64, S
Poly(phenyl acetylene)	THF	632.8	0.2864	25		65
Epoxy resin	THF	632.8	0.187	23		20
Phenoxy resin	THF	632.8	0.1773			26
Polycarbonate	THF	632.8	0.177			20
		632.8	0.1855			20
Polyphosphazene I	THF	632.8	0.160	30		66, T
II		632.8	0.145	30		66, T
III		632.8	−0.029	30		66, T
IV		632.8	−0.023	25	1.0	67, T
	acetone	632.8	0.019	25	5.1	67, T
	cyclohexanone	632.8	−0.053	40	6.7	67, T
Poly(ether sulfone)	DMF	632.8	0.188	25		4
Polyimide	DMF	632.8	0.208	25		4
	N-methylpyrrolidone	632.8	0.132			68, U
PEEK						
(MW 14300–22600)	conc. H_2SO_4	632.8	0.376		4.3–2.0	69, V
(MW 28300–56500)		632.8	0.419		1.8–2.7	69, V
Poly(phenylene sulfide)						
	1-chloronaphthalene	632.8	0.167	220		70
Poly(vinyl butyral)	HFIP	632.8	0.189	25		30, J
	THF	632.8	0.089			30, J

Sample	Solvent	Wave-length (nm)	dn/dc (ml/g)	Temp (°C)	A_2 (× 10^3)	Ref, Remark
Poly(2-vinylpyridine)	benzene	546	0.075	25		71
Poly(4-vinylpyridine)	DMF	436	0.160	25		10
	chloroform	436	0.150	25		10
Poly(vinylpyrrolidone)	DMF	632.8	0.095	23		20
	chloroform	546	0.108			10
		436	0.108			10
	water	546	0.185			10
		436	0.185			10
6 FDA/p-PDA	*N*-methylpyrrolidone	632.8	0.107			72, W
6 FDA/MDA		632.8	0.132			72, W
BTDA/DDS		632.8	0.145			72, W
Methylated humic acid	THF	632.8	0.22		−0.042	73
Benzylated humic acid	THF	632.8	0.19		−0.041	73
Hydroxyethyl cellulose acetate	THF	488	0.0734	25		74
Poly(vinyl acetate)	THF	632.8	0.0517	25		11
		632.8	0.0471	26		20, X
		632.8	0.0402	26		20, Y
	ethyl acetate	632.8	0.077	23		20
	acetone	546	0.104	25		10
		436	0.104	25		10
	MEK	546	0.080	25		10
		436	0.083	25		10
Poly(vinyl alcohol)	0.05 M $NaNO_3$	632.8	0.1501	27		75
	0.05 M $NaNO_3$	632.8	0.1429	27		75, Z
		632.8	0.1501	25		76, a
		632.8	0.1429	25		76, b
	0.1 M KH_2PO_4	632.8	0.1534	23		20
	water	546	0.164	30		10
		436	0.168	30		10
Polyacrylamide	water	632.8	0.1735			77
		632.8	0.142	23		20, C
		632.8	0.1829	25		20, d
		546	0.163			10
	1 M NaCl	632.8	0.1587	25		20, e
	1 M NaCl	632.8	0.1556	25		20, f
	0.1 M Na_2HPO_4	632.8	0.163			78
	0.02 M Na_2SO_4	632.8	0.176	23		79, g
			0.174	23		79, h
Poly(acrylamide-co- -sodium acrylate)	1 M NaCl	632.8	0.165			80
	0.1 M NaH_2PO_4 + 0.3 M Na_2SO_4, pH 2.2	632.8	0.16			81
(70/30)	0.1 M $NaNO_3$	632.8	0.165			20
Sodium polyacrylate	0.3 M NaCl	488	0.159	25		82
	0.1 M NaCl	632.8	0.175	23		20
	water	632.8	0.1463	23		20, i

Sample	Solvent	Wave-length (nm)	d*n*/d*c* (ml/g)	Temp (°C)	A_2 (× 10^3)	Ref, Remark
Poly(acrylic acid)	aqueous NaCl	436	0.158	30		10
	THF	632.8	0.0738	23		20
		632.8	0.0994	23		20
	0.2 N HCl	632.8	0.133			83
Poly(allyl amine)	0.1 % TFA/	632.8	0.2034	35		84, j
	0.20 M $NaNO_3$	632.8	0.1950	35		84, k
Poly(vinyl amine)	0.1 % TFA/0.20 M $NaNO_3$	632.8	0.1751	35		84, m
Branched polyethyleneimine						
	water	632.8	0.210	35		85
	1 M NaCl	632.8	0.195	35		85
	methanol	632.8	0.225	35		85
Poly(ethylene oxide)	methanol	632.8	0.145	25		86
Dextran	water	632.8	0.147			87, n
	water	632.8	0.150	25		88
	0.1 M KNO_3	632.8	0.150	25		88
	0.05 M KH_2PO_4 (pH 7)	632.8	0.137			87, n
	0.15 M NaCl	632.8	0.145			89
	0.05 M K_2HPO_4 (pH 7)	632.8	0.1378		0.41	90
	water	546	0.148			91
	0.5 M NaCl	546	0.145			91
	water	436	0.150			91
	0.5 M NaCl	436	0.148			91
	0.2 M $NaNO_3$	514.5	0.147	RT		92
DEAE-dextran	0.8 M $NaNO_3$	632.8	0.150	20		93, o
Pullulan	0.2 M $NaNO_3$	514.5	0.147	RT		92
	0.15 M NaCl	632.8	0.145			89
	water	488	0.148	25		94
Polysaccharide	water	632.8	0.146			95
	0.1 M NaCl	632.8	0.143			96, p
(anionic)	0.1 M NaCl	632.8	0.143			96, q
Sodium hyaluronate	0.1 M NH_4NO_3	632.8	0.155		3	97
Sodium heparin	0.1 M NaCl	632.8	0.163	23		98, r
Sodium poly(styrene sulfonate)						
	water	632.8	0.17			99
(0.9 meq/g)	0.05 M NaCl	546	0.212	25		10
(1.8 meq/g)		546	0.207	25		10
(2.0 meq/g)	0.10 M NaCl	546	0.195	25		10
(2.7 meq/g)		546	0.205	25		10
	0.1 M NaNO	632.8	0.195			100
Poly(sulfoalkyl methacrylate)						
(ethyl)	0.1 M NaCl	532	0.135	25		101
(hexyl)		532	0.140	25		101
(decyl)		532	0.170	25		101
(octyl)		532	0.149	25		101
Hydroxyethyl-cellulose	0.1 M NaCl	632.8	0.159			102

Sample	Solvent	Wave-length (nm)	d*n*/d*c* (ml/g)	Temp (°C)	A_2 ($\times 10^3$)	Ref, Remark
Carboxylmethyl-cellulose	0.1 M NaCl	632.8	0.147			102
PNaAMPS	water	632.8	0.20			99, s
BSA	water	632.8	0.187	25		103, t
BSA-SDS	water	632.8	0.376	25		103, u
Ovalbumin-SDS	water	632.8	0.374	25		103
Ribonuclease-SDS	water	632.8	0.398	25		103
Ribonuclease A	75% A – 25% B	632.8	0.172		7.5×10^{-2}	104, v
Lysozyme	65% A – 35% B	632.8	0.182		-5.39×10^{-2}	104, v
BSA	58% A – 42% B	632.8	0.172		7.49×10^{-2}	104, v
Bovin alkaline phosphatase	50% A – 50% B	632.8	0.142		0.854	105, w
	60% A – 40% B	632.8	0.1188		−9.81	105, w
	70% A – 30% B		0.0956		−163	106, w
	70% A – 30% B	632.8	0.0956		−163	105, w
Bovin thyroglobulin (MW 669 000)	0.10 M Na_2SO_4	632.8	0.165	20		106, x
Serum albumin (MW 68 000)	0.10 M Na_2SO_4	632.8	0.165	20		106, x
Chymotrypsinogen A (MW 25 900)	0.10 Na_2SO_4	632.8	0.170	20		106, x
DNA	water	632.8	0.166			99
DNA fragment	0.1 M NaCl	632.8	0.168	25	0.6	107

Remarks. A: $dn/dc = 0.108 - 21.0/M_n$; B: $dn/dc = 0.106 - 12.4/M_n$; C: polystyrene MW-$(dn/dc)_{toluene}$-$(dn/dc)_{MEK}$, 1.8×10^6–0.109–0.216, 97 200–0.107–0.214, 51 000–0.106–0.212, 19 800–0.103–0.213, 10 300–0.102–0.209, 4800–0.101–0.210, 2030–0.0985–0.208, polyethylene/trichlorobenzene $dn/dc = -(0.1023 + (20.4/M))$; D: $(dn/dc)_{670} = 0.9783\ (dn/dc)_{632.8}$, $(dn/dc)_{670} = 0.1804 - 9.149/M$; E: methylene chloride; F: DMF – *N,N*-dimethylformamide; G: 60% MMA diblock; H: PS/THF, $A_2 = 0.01\,M_w^{-0.25}$, PMMA/THF, $A_2 = 0.012\,M_w^{-0.29}$; J: vinyl alcohol content ranging from 11 to 23%, hexafluoroisopropanol; K: $M_w = 6160$; L: $dn/dc = 0.111 - 15.3/M_n$ (36% *cis*-1,4, 57% *trans*-1,4, 7% 1,2-); M: $M_w = 6100$ (0.0974 for $M_w = 3300$); N: ethylene-co-octene-1; O: ethylene-co-butene-1; P: pentafluorophenol; Q: dichloroacetic acid; R: M_w – dn/dc, 35 500 – 0.1417, 46 200 – 0.1523, 63 100 – 0.1477, 72 400 – 0.1469; S: poly(methyl vinyl ether-co-maleic anhydride), hydroxymethyl aminomethane, pH 9; T: I – poly (diphenoxy)phosphazene, II – poly(aryloxy)phosphazene, III – poly(fluoroalkoxy)phosphazene, IV – poly[bis(trifluoroethoxy) phosphazene]; U: 6F-dianhydride-1,4-phenylenediamine $dn/dc = 0.107$, 6F-dianhydride-4,4-methylenediamine $dn/dc = 0.132$, benzophenone-3,3′,4,4′-diaminodiphenylsulfone $dn/dc = 0.145$; V: poly(arylether ether ketone); W: 6FDA – 6F-dianhydride, MDA – 4,4′-methylenediamine, DDS – 4,4′-diamine diphenyl sulfone, BTDA – benzophenone-3,3′,4,4′-tetracarboxylic dianhydride, PDA – 1,4-phenylenediamine; X: $M_w = 4.66 \times 10^6$; Y: $M_w = 1.07 \times 10^5$; Z: 88% degree of hydrolysis; a: 98% hydrolysis; b: 88% hydrolysis; c: $M_w = 5.5 \times 10^6$; d: $M_w = 3.69 \times 10^6$; e: $M_w = 7.72 \times 10^5$; f: $M_w = 8 \times 10^6$; g: $5.32 \times 10^5\ M_w$; h: $2.18 \times 10^6\ M_w$; i: $M_w = 28000$; j: TFA – trifluoroacetic acid, $M_w = 3.61 \times 10^4$; k: $M_w = 1.06 \times 10^5$; m: $M_w = 2.50 \times 10^5$; n: $M_w = 6.82 \times 10^4$; o: DEAE – 2-diethylaminoethyl; p: Schizophyllan (neutral polysaccharide); q: xanthan (anionic heteropolysaccharide); r: 0.01 M NaCl – 0.142 – A_2 = 16, 0.1 M NaCl – 0.137 – 5.8, 0.5 M NaCl – 0.126 – 2.52; 1 M NaCl – 0.121 – 1.52; s: poly(sodium-2-(acrylamide)-2-methyl propanesulfonate); t: bovine serum albumin; u: sodium dodecyl sulfonate; v: A – 0.15 % TFA in water, B – 0.15 % TFA in 95 % acetonitrile; w: A – 2.5 M ammonium sulfate in 0.5M ammonium acetate (pH 6.0), B – 0.5M ammonium acetate (pH 6.0); x: in 0.02 M Na_2HPO_4 – NaH_2PO_4 at pH 6.9.

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